Combining Unsupervised Image Segmentation and Semi-Supervised Video Object Segmentation

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Abstract

While Video Object Segmentation has improved substantially, it still requires carefully procured videos and annotations for each frame. Data labelling is expensive and serves as a bottleneck for VOS. Semi-supervised VOS, in which an annotated mask is only required for the first frame, is an alternative to fully-supervised VOS, but mask annotations are still much more difficult to generate compared to, for example, image classification labels. We explore concatenating unsupervised image segmentation and semi-supervised VOS to produce a fully-unsupervised VOS method. The mask output of the unsupervised image segmentation model is used as the first frame mask for the semi-supervised VOS task; in this way, we remove the need for any annotations at all. Our method produces reasonable results on the DAVIS 2016 dataset and proves to be a promising area for future exploration.

1. Introduction

Video Object Segmentation (VOS) concerns the task of labelling each pixel in a sequence of video frames as foreground or background. VOS has many applications ranging from engineering (object detection, autonomous vehicles, and robotics) to art (video editing and animation). Image segmentation has seen many breakthroughs that have extremely accurate performance, such as Mask R-CNN [6]. Adding a time dimension to the task of image segmentation, however, proves to be a much more difficult task. VOS differs from semantic segmentation, in which the video pixels are clustered not into two classes, but into many semantic regions of the frames.

The task of VOS is a step-up from image segmentation in multiple ways. Inherently, the time dimension of videos adds a new level of complexity. However, the segmentation task itself is not the only hurdle impeding progress in VOS. There are few well-maintained and curated VOS datasets due to the high cost and inherent difficulty in generating mask annotations. DAVIS 2016 [13] provided a respectable number of fully-annotated videos and benchmarks to evaluate model performance on the dataset. DAVIS 2017 [14] extended the 2016 dataset and also introduced mask annotations for Video Instance Segmentation, in which multiple instances of objects are detected in a single video. In 2019, 4000 high-resolution videos with mask annotations for both video object and instance segmentation were released in YoutubeVOS [22]. DAVIS and YoutubeVOS are the major datasets in use for VOS, but they pale in comparison to CIFAR-10 and ImageNet. Until a cheaper and faster pipeline for annotating videos is developed, video segmentation has to make do with less data. An additional issue arises with evaluation; methods for analyzing the performance of VOS models without having ground truth masks have yet to be refined.

A mitigation for this problem is *semi-supervised video object segmentation*, in which an annotation is provided for only the first frame of the video. This is a popular approach with VOS because it requires a much lower number of manual annotations. Methods such as RANet [17] and STCNN [21] take advantage of the first mask to produce compelling results. However, even semi-supervised VOS does not have access to large datasets, and it is difficult to apply in a practical situation, as it requires the input of the first frame's mask.

One interesting property of videos are that they are simply images with an extra time dimension. While audio signals can be broken up into phonemes and time series into individual data points, those types of data do not have the structure and information that images carry. Even by themselves, images provide a plethora of data to analyze and process. Thus, it makes sense to leverage this structure. Even though semi-supervised VOS requires an annotation of the first frame, there is no requirement that it is manually generated. We can take advantage of existing image segmentation models to do this task for us, and if the result is comparable to a human-generated mask, then semi-supervised VOS should perform just as well. In a way, we are performing self-supervised VOS, with an unsupervised image segmentation model providing the supervision.

The rest of this paper is structured as follows: we discuss related works in Section 2, explain our methods in Section 3, present our experiments in Section 4, and summarize our conclusions in Section 5.

2. Related Works

Image Segmentation The task of image segmentation has been relatively thoroughly researched. Many supervised algorithms have achieved superb performance on existing datasets, including Mask R-CNN by He *et al*. [6], U-Net by Ronneberger, Fischer, and Brox [15], Fully Convolutional Networks by Long, Shelhamer, and Darrell [10], and DeepLabv3 from Chen *et al*. [2].

Unsupervised image segmentation is less explored. W-Net extends U-Net with an encoder-decoder architecture and a Soft Normalized Cuts loss [20]. Kanezaki used back-propagation in images to iteratively group pixels with similar features into the same class [7]. ReDO, by Chen, Artlères, and Denoyer, generatively builds models by creating segmentations, filling in pixel colors, and assembling different regions together [3]. In LOCUS, Winn and Jojic mix low-level features such as color and edges with high-level features such as shape and pose to iteratively identify objects in images [19]. Similarly, in GLGOV, Yan *et al*. combine those low-level features with Gestalt principles to generate image segmentations. Ji, Henriques, and Vedaldi took an information-theoretic approach an proposed Invariant Information Clustering to perturb images, generate segmentations of the pertubations, and, in a semi-supervised manner, predict class labels from the segmentations and maximize mutual information of the labels. Unsupervised models generally achieve lower performance than supervised models, but they are rapidly catching up and levelling out the playing field.

Semi-Supervised VOS Many semi-supervised VOS models have achieved very high performance on the DAVIS and YoutubeVOS datasets. These methods often have a tradeoff between sharp segmentations and temporal consistency: either the masks jitter between frames, or they are noisy and include parts of the background. Newer VOS models often offer novel ideas to mollify these issues.

Wang *et al*. proposed RANet, which uses a ranking attention module to select similarity maps for generating masks [17]. Xu *et al*. propose Spatiotemporal CNN, with a spatial branch and a temporal branch, to produce segmentations for each frame while encouraging temporal consistency [21]. Li and Loy use a recurrent network to track objects even when their appearance has changed between frames [9]. Cheng *et al*. tackle the same issue by tracking parts of objects instead of entire objects [4]. Ochs, Malik, and Brox exploit the ”common fate” Gestalt principle to group video pixels together [12].

Unsupervised VOS Unsupervised VOS algorithms are rapidly improving and achieving performance close to that of semi-supervised VOS algorithms. Similar to [12], MAT-Net uses a Motion-Attentive Transition block to combine motion and appearance to track objects [24]. Yang *et al*. leverages long-term temporal dependencies in An-Diff to establish correspondences between video frames and an anchor image [23]. Lu *et al*. use co-attention layers with short stacks of video frames to learn relationships between appearance and motion [11]. Wang *et al*. incorporate eye-tracking data for a more biological perspective of object tracking [16]. While other works have explored segmenting each frame of a video and smoothing the result with continuity priors, they depend on interconnected models trained together specifically for unsupervised VOS. No other works have just thrown together separate models for unsupervised image segmentation and semi-supervised VOS to exploit the image-video relationship.

3. Methods

In this paper, we select RANet [17] as our semi-supervised VOS model, and we experiment with both W-Net [20] and ReDO [3] for our unsupervised image segmentation models. In addition, we compare the results of VOS when an unsupervised image segmentation model is used, as opposed to a supervised image segmentation model.

3.1. RANet

RANet is one of the leading semi-supervised VOS models for the DAVIS 2016 datasets. It consists of an encoder-decoder architecture for feature mapping with a novel *Ranking Attention Module* (RAM) in between to extract the most salient features for segmentation.

The first module of RANet uses a Siamese encoder to generate features from the first image of the video, as well as the image at time t. If we denote the features of the first frame and current frame by $I^{(1)}, I^{(t)} \in \mathbb{R}^{C\times H\times W}$, where the frames have height H, width W, and number of features C, then we generate a correlation map $S \in \mathbb{R}^{HW\times H\times W}$. The RAM takes as input the first frame mask to filter out background pixels from the similarity maps. Then, a two-layer fully-connected network is used to learn a ranking score r_j for each of the HW similarity maps $S_j \in \mathbb{R}^{H\times W}$. Finally, the similarity maps are ranked by their ranking scores, and the top N maps are retained. These similarity maps are then merged with the mask at frame $t-1$ and fed into a decoder network to generate the mask at time t.

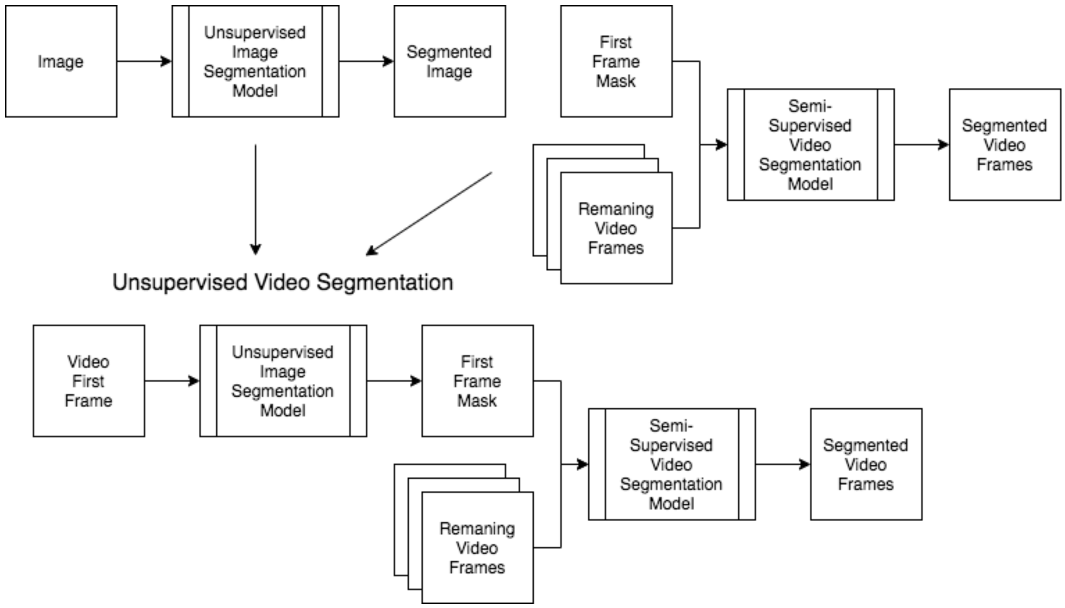

Figure 1. The first frame of the video is fed into the unsupervised image segmentation model. The output mask is then input into the semi-supervised video segmentation model along with the remaining frames of the video. The resulting video object segmentation model is fully unsupervised.

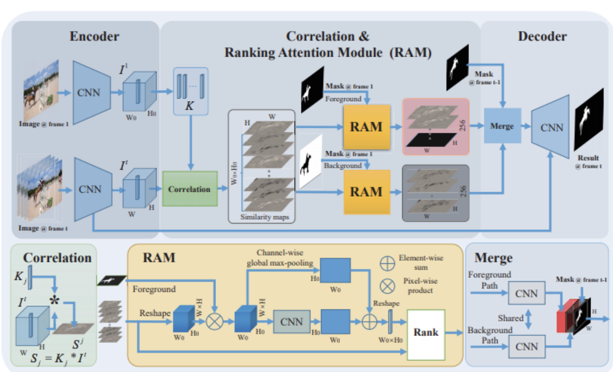

Figure 2. The RANet architecture. Figure from [17].

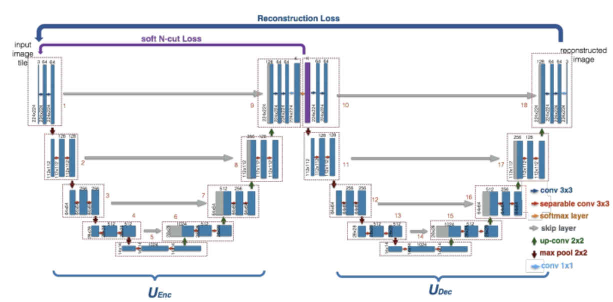

Figure 3. The W-Net architecture. Figure from [20].

We use a pre-trained RANet with $N = 256$. The Siamese encoders are built from a ResNet-101 network pre-trained on ImageNet, and the decoder is a pyramid network.

3.2. W-Net

The first unsupervised image segmentation model that we experimented with is W-Net, an extension of U-Net. It consists of two U-Net architectures stacked together in an encoder-decoder architecture. The encoder takes in the input image $I \in \mathbb{R}^{H \times W \times C}$ and generates a segmentation with K classes $I_S \in \mathbb{R}^{H \times W \times K}$. Then, this segmentation is fed back into the decoder, which outputs a reconstruction of the original image, $I_R \in \mathbb{R}^{H \times W \times C}$.

The model is trained with a dual loss. An ℓ_2 reconstruction loss is imposed on the reconstructed image and the original image. In addition, a differentiable soft normalized-cuts loss is used to reduce noise in the segmentation output from the encoder.

The architecture of the U-Nets in the W-Net diverges slightly from the original U-Net paper. It uses depth-wise separable convolutions composed from an independent depthwise and pointwise convolution in order to separate spatial correlations from cross-channel correlations.

The soft normalized cut is defined as follows, where A_k is the set of pixels in segment k, V is the set of all pixels in the image, and w is a weight function that defines the weight between two pixels:

$$\begin{aligned}J_{soft-Ncut}(V,K) &= \sum_{k=1}^{K} \frac{cut(A_k, V - A_k)}{assoc(A_k, V)} \\ &= K - \sum_{k=1}^{K} \frac{assoc(A_k, A_k)}{assoc(A_k, V)} \\ &= K - \sum_{k=1}^{K} \frac{\sum_{u\in V, v\in V} w(u,v)p(u=A_k)p(v=A_k)}{\sum_{u\in A_k, t\in V} w(u,t)p(u=A_k)} \\ &= K - \sum_{k=1}^{K} \frac{\sum_{u\in V} p(u=A_k)\sum_{v\in V} w(u,v)p(v=A_k)}{\sum_{u\in V} p(u=A_k)\sum_{t\in V} w(u,t)}\end{aligned} \tag{1}$$

The weight w_{ij} between pixels i and j is defined as follows, where $X(i)$ is the spatial location of the pixel, $F(i)$ is the pixel value, and σ_I, σ_X, and r are tuned parameters.

$$w_{ij} = \begin{cases} exp(-(\frac{\|F(i)-F(j)\|_2^2}{\sigma_I^2} + \frac{\|X(i)-X(j)\|_2^2}{\sigma_X^2})) & \text{if } \|X(i)-X(j)\|_2 < r, \\ 0 & \text{o.w.} \end{cases} \tag{2}$$

Finally, the output segmentation from the encoder is post-processed with a fully connected Conditional Random Field [8], and segments are merged through hierarchical segmentation [1]. In this paper, we post-process with fully connected CRF, but not hierarchical segmentation. We train the W-Net using the PASCAL VOC 2012 dataset [5].

3.3. ReDO

In contrast to other approaches that classify pixels in an image, ReDO uses a generative approach to segment images by reconstruction. Similar to how a human might draw an image, ReDO's model starts by producing a segmentation before filling in the pixels of each region in the segmentation with colors. Finally, the regions are combined into a single image.

More formally, we take a function $F(I;n) \rightarrow (M_1,\dots,M_n)$ that accepts an image $I \in \mathbb{R}^{H\times W\times C}$ and a parameter n for the number of regions in the image, including the background. This learned function F is the segmentation function. Next, we consider n different generators $G_i(M^{(i)}, z_i)$, each of which takes in a mask and a vector sampled from a prior $p(z)$, and generates the pixel values $V^{(i)}$ for region i. Finally, the masks and generated pixel values are combined into the final reconstructed image $I_R = \sum_{i=1}^{n} M^{(i)} \odot V^{(i)}$.

The model is trained with an adversarial loss. To prevent the segmentation function F from trivially classifying all of the pixels in the image as one region, only one mask is redrawn for each sample; this redrawn region is combined with the original image and classified by the discriminator. In addition, a mutual information loss is introduced to encourage the redrawn image to contain information about the prior vector z_i. Once training is complete, the segmentation function F can be used to segment images, similar to the encoder from W-Net.

For this paper, we use a ReDO segmentation model pretrained on the CUB dataset [18]. We were unable to finetune the model on the PASCAL VOC 2012 dataset, but the resulting image segmentations were fairly reasonable.

3.4. Metrics

The DAVIS dataset uses three main metrics to evaluate model performance on the VOS task [13]. The first is the Jaccard index, which directly measures the similarity between the predicted mask M and the ground truth mask G: $J(M,G) = |\frac{M\cap G}{M\cup G}|$. The second is the F-measure, which evaluates the accuracy of the contour points $c(M)$ and $c(G)$ of the mask and ground truth. With precision P_c and recall R_c, the F-measure is defined as $\frac{2P_cR_c}{P_c+R_c}$. For both the Jaccard index and the F-measure, DAVIS computes the mean, recall, and decay, although we focus on the mean in this paper. Perazzi *et al*. also define a third measure of temporal stability, but we do not include it here.

4. Experiments

We tested three combinations of image segmentation and semi-supervised VOS models: DeepLabv3-RANet, W-Net-RANet, and ReDO-RANet. The DeepLabv3 model used was the pre-trained model from torchvision. Note that unlike the other models benchmarked with DAVIS 2016, we crop photos to a size of 224×224 in order for model compatibility. As a result, our results are not fully representative of the DAVIS dataset.

4.1. Unsupervised Image Segmentation

Surprisingly, one of the main obstacles was the unsupervised image segmentation. Although there is much literature on the topic, the code for this field is lacking, so we implemented our own version of W-Net. Compared to the supervised image segmentation model, W-Net results were far noisier (Figure 4). DeepLabv3 segmentations exhbited some inaccuracy, but for the most part fit the image foreground well. ReDO generally deciphered the outline of the object in the image, but would sometimes include small patches of surrounding background or exclude small patches of the object.

4.2. Combined Models

Below, we include samples of various portions of the test sequences (Figure 5). As expected, because the first-frame masks produced by DeepLabv3 match the foreground very well, the subsequent masks produced by RANet also hug

Figure 4. Image segmentation results for DeepLabv3 (top), W-Net (middle), and ReDO (bottom).

Sequence	J	D-J	W-J	R-J	F	D-F	W-F	R-F
blackswan	94.4	89.5	8.2	46.8	96.7	94.7	25.0	61.6
bmx-trees	71.0	0.3	12.1	15.5	91.3	1.6	49.6	36.7
breakdance	82.2	25.9	13.0	32.2	79.4	27.4	23.3	34.2
camel	92.2	77.8	27.2	30.4	95.0	84.5	38.7	43.5
car-roundabout	94.9	91.9	2.7	60.2	92.7	84.0	22.7	46.6
car-shadow	95.9	91.9	15.9	71.3	98.7	94.6	16.3	57.3
cows	94.7	89.6	2.9	38.6	97.5	95.0	13.0	27.8
dance-twirl	86.6	30.1	0.3	23.1	90.6	41.2	8.8	24.2
dog	95.6	89.0	62.4	63.2	97.6	89.1	49.7	61.9
drift-chicane	???	6.6	5.7	0.0	91.2	18.4	17.5	0.0
drift-straight	91.7	18.2	3.9	39.1	93.5	27.2	9.6	21.0
goat	88.8	86.7	0.3	5.6	90.4	87.2	8.6	16.9
horsejump-high	88.5	67.9	1.3	43.6	95.3	82.3	23.1	48.6
kite-surf	69.0	67.4	1.7	15.0	69.8	82.0	16.0	43.9
libby	86.0	70.5	11.3	58.0	95.1	88.3	42.6	75.2
motocross-jump	86.7	28.3	2.3	23.5	79.2	38.2	22.5	35.7
paragliding-launch	64.1	81.5	0.7	75.5	28.5	89.6	21.5	81.4
parkour	93.3	83.8	2.8	56.3	96.3	94.0	23.7	65.9
scooter-black	85.8	48.4	0.2	32.8	79.1	48.0	11.3	29.5
soapbox	89.4	3.7	22.8	12.6	87.5	17.1	42.7	33.8
Average	82.4	57.5	9.9	37.2	80.7	64.2	24.3	42.3

Table 1. The mean Jaccard (J) index and F-measures (F) of each method for each test sequence. We denote DeepLabv3-RANet by D, W-Net-RANet by W, and ReDO-RANet by R. State-of-the-art J and F are listed in the first and fourth columns, with question marks ??? denoting a missing value.

the object tightly. Because of RANet's temporal dependencies, the masks are smooth from one frame to the next and cleanly follow the foreground object. The benchmark results for DeepLabv3-RANet have high variance, but on certain sequences, they are very high performance and competitive with state-of-the-art results. Human annotations are not required for semi-supervised learning; as shown by the DeepLabv3-RANet, the masks can be provided by separately-trained models.

The VOS results from W-net are respectable. While most sequences have a low Jaccard index, even with such a poorly-trained model, we achieve high performance on the dog sequence, and several other sequences are acceptable with respect to the F-measure. While the results are noisy, they are able to isolate the bike and the dog. In the case of the bike, the grass is also mistakenly identified as part of the biker, although as the tree begins sweeping into frame, the grass is detached from the biker. The mask for the dog begins with some noisy patches before converging onto the dog. Although there are small patches of grass included, the model is able to capture the entire dog with the mask. Furthermore, the trend is consistent across different sequences. Although masks generated by W-Net for the first frame are typically noisy, RANet filters out excess patches as the frames progress. It seems likely that when the RANet decoder processes the previous mask, it filters out noise that does not generate much signal when combined with the similarity maps. Gradually, noise patches are removed from the output.

As expected, because ReDO produces much cleaner masks compared to W-Net, ReDO-RANet also generates better masks than W-Net. In (Figure 5), the model only misses patches of the car once it enters the sunlight and drastically changes color. In the bottom row, libby weaves in and out of the trees, but ReDO-RANet is able to follow it, even excluding the occluding trunks. Although the masks created by ReDO are not perfect, they are a huge step up from those of W-Net, and the difference is clear between the unsupervised VOS results; moreover, the same trend is present as before, where patches of noise diminish in later frames. The benchmark results for ReDO-RANet are much better than those of W-Net-RANet. Multiple models reach a Jaccard index of over 60, indicating very promising results.

Note that in several cases, such as paragliding-launch and kite-surf, the joint models outperform state-of-the-art models. There are several possible explanations for this. First, the center cropped images change the exact meanings of the metrics, so they are not directly comparable. Second, although DeepLabv3-RANet beats state-of-the-art for the paragliding-launch sequence, it has high variance, and overall is not consistent.

5. Conclusion

While our results are not state-of-the-art, they hold promise for an avenue of research. They introduce a new approach to VOS that makes good use of the structural relationship between images and videos. Without fine-tuning or extra adjustments or hacking, the models achieve satisfactory performance. In fact, our video segmentations with DeepLabv3-RANet indicate that if the unsupervised image segmentation models are improved to produce better quality segmentations more consistently, then the combined unsupervised models will be much more competitive with semi-

Figure 5. Each row depicts ten consecutive frames of a video sequence, with the predicted mask colored in by cyan or magenta. The top two rows (blackswan, parkour) are from DeepLabv3-RANet, the middle two rows (bmx-trees, dog) are from W-Net-RANet, and the bottom two rows (car-shadow, libby) are from ReDO-RANet.

supervised VOS models.

With the limited time frame of this project, we were unable to fully tune our unsupervised image segmentation model. After building it from scratch and exhaustively iterating through hyperparameters to no avail, we had to fall back to an implemented model, even though it was tuned on the wrong dataset. With more time, we would have fine-tuned ReDO on PASCAL and DAVIS images to improve its generalization.

In the future, we could explore fine-tuning the joint model to improve performance. Although this detracts from the beauty in the disjoint nature of the model, fine-tuning would likely make the model even more competitive with current state-of-the-art methods. In addition, it would be interesting to explore variations in combining the image segmentation and semi-supervised VOS models. For example, we could output a mask from the unsupervised image segmentation model for every frame, not just the first frame. This would provide more supervision for the semi-supervised VOS model and likely be beneficial.

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